Reading and Making Graphs

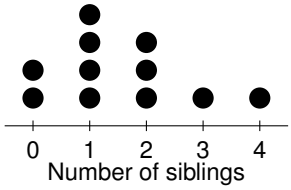
Name _____

Frequency means how many times something happens. These three tools all show frequency.

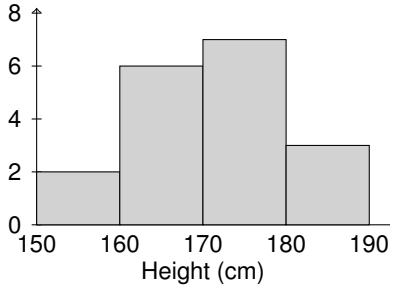
Frequency table

Siblings	Tally	Frequency
0	II	2
1	IIII	4
2	III	3
3	I	1
4	I	1

Dot plot



Histogram



Each **value** gets a row.	Each **value** gets a spot on a number line.	Values are put in **groups** (bins).
Make a tally mark for each data point, then count.	**One dot = one data point.** Stack the dots.	Bars **touch**. No gaps between groups.
Frequency column = how many.	Height of the stack = how many.	Bar height = how many in that group.

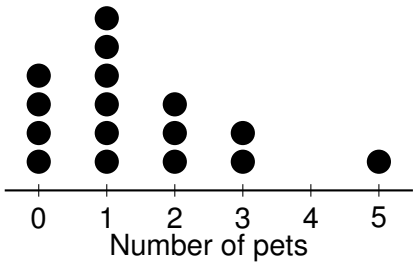
Histogram rule: If a number is **on a line** between two bars, it goes in the bar on the **right**. Example: 160 goes in the 160 to 170 bar.

The frequency table and dot plot above show the **same data**. Check: 2 + 4 + 3 + 1 + 1 = 11 students.

Every time you read a graph: read the title, read both axis labels, then check the scale.

Work with a partner. Compare answers when you finish each problem.

1. This dot plot shows how many pets some students have.



(a) How many students have exactly 2 pets?

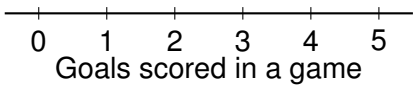
(b) How many students are in the data?

(c) How many students have more than 2 pets?

(d) What number of pets is most common?

2. A soccer team scored these goals in 12 games. Make a dot plot.

2, 0, 1, 3, 1, 1, 4, 2, 0, 1, 2, 1



3. 20 students said how many siblings they have. Make a frequency table.

1, 2, 0, 3, 1, 1, 2, 4, 0, 2, 1, 3, 2, 1, 0, 2, 1, 5, 2, 1

Siblings	Tally	Frequency
0		
1		
2		
3		
4		
5		

(a) How many students have 2 or more siblings?

(b) How can you use your table to check that you counted everyone?

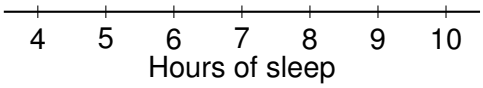
4. This frequency table shows how many hours some students slept last night.

Hours of sleep	Frequency
5	2
6	4
7	6
8	5
9	3

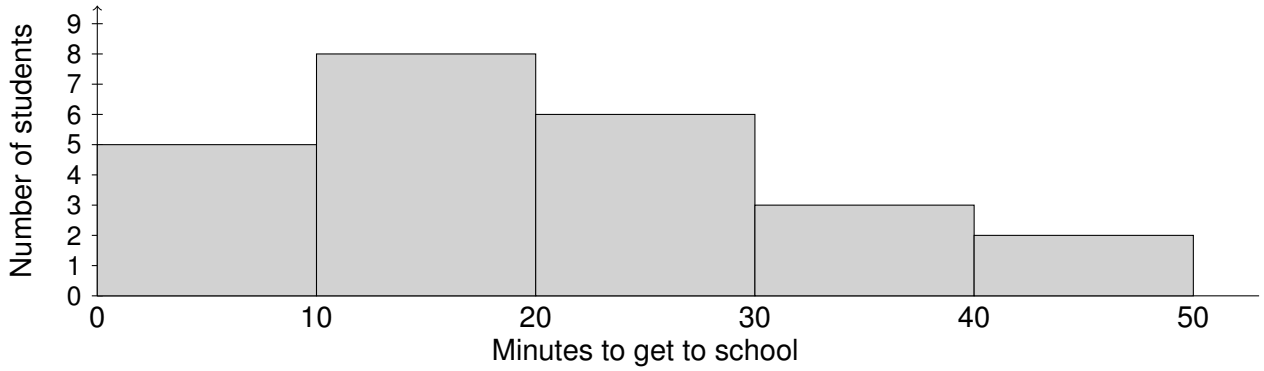
(a) How many students are in the data?

(b) How many students slept less than 7 hours?

(c) Make a dot plot of the same data.



5. This histogram shows how many minutes some students take to get to school.



(a) How many students take from 20 to 30 minutes?

(b) How many students take less than 20 minutes?

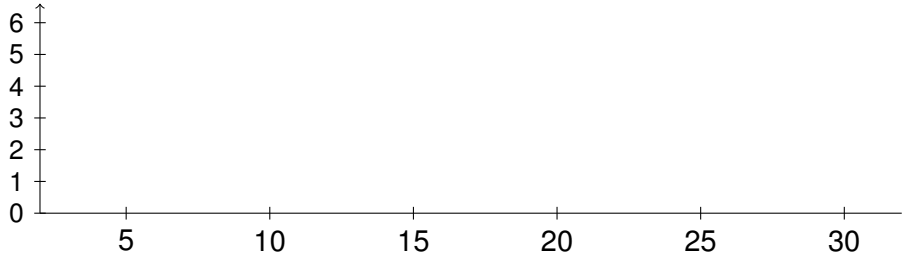
(c) How many students are in the data?

(d) Can you tell from this graph how many students take exactly 15 minutes? Explain.

6. A basketball player scored these points in 15 games.

12, 8, 15, 21, 9, 14, 18, 10, 25, 13, 7, 16, 20, 11, 19

(a) Make a frequency table. Count how many games are in each group. Remember the histogram rule.

Points	Frequency
5 to 10	
10 to 15	
15 to 20	
20 to 25	
25 to 30	

(b) Make a histogram. Label both axes.